Math of Origami Constructions

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1 Introduction

Origami is the art of folding paper to create shapes and objects, originating in Japan. On the surface, origami may seem like another creative hobby, sometimes simple and sometimes very intricate. But when you unfold and look within each crease, the mathematical principles become more evident.

Origami techniques involve all kinds of math. For instance, dividing segments, forming diagonals, and keeping symmetry for constructions are all patterns commonly seen in even the most basic constructions. Furthermore, origami can be used for different kinds of problem solving, such as creating parabolas and even solving cubic equations.

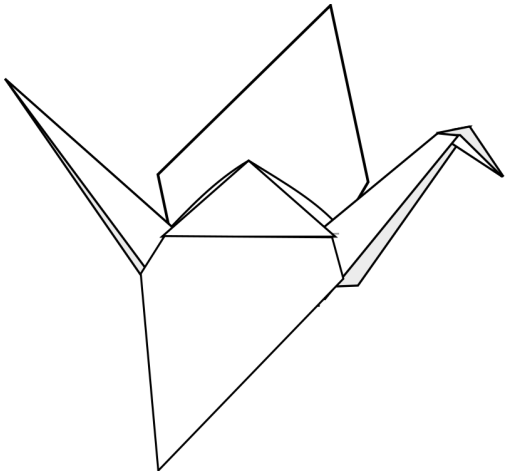


Figure 1: Origami Crane

2 Ruler and Compass Constructions

Before we get into what we can do with origami, we first have to understand lengths formed from rulers and compasses. These numbers are called constructible numbers. But what actually counts as a constructible number? Constructible numbers represent the set of numbers that can also express the length of a segment formed using a straight edge or compass. To put it simply, given a unit length (a construction with the length of one unit), constructible numbers can be formed by drawing segments in between points, drawing circles with a given center and point on its circumference, and finding the intersections of segments and circles.

The use of constructible numbers, not a formalized concept at the time, originated from Ancient Greek geometry. These numbers were fundamental for attempting to solve problems like trisecting an angle, doubling a cube, or squaring a circle. In fact, constructible numbers were implicitly foundational (not algebraically formalized) to Euclid's math, and Euclid would use these concepts to work with ratios of lengths, such as the ratio of a diagonal of a square to its side length.

We can perform basic arithmetic such as adding, multiplying, and even taking the square root of existing constructible numbers to form other constructible numbers. Note that it is assumed we are able to construct a unit length.

1. To copy a line segment, you can use a compass to measure the length of a line segment, and then place the compass on the paper and use a ruler to construct the line segment.

2. You can add line segments by measuring their lengths, marking each distance onto the paper (where the start point of each line segment is the endpoint of the last), and then using a ruler to draw the line segment such that it passes through all of the points of the markings. Subtraction can be done the same way, except the direction of the second segment is opposite to the direction of the first.

3. The product of two constructible numbers, a and b, can be found by drawing them to form an acute angle, drawing a line from the endpoint of segment a to b and drawing another line from the endpoint of b to the extension of a, parallel to the previous line, and taking the length from the shared endpoint to the endpoint of the second constructed line.

4. To find the quotient of two constructible numbers, with the same setup of segments of lengths a and b, construct a line from the endpoint of b to a and then construct a line parallel to the previous one where the endpoint on line b is 1 unit away to the shared endpoint of a and b. The length from the shared endpoint to the second line, at a closer distance to the shared endpoint than the first, represents $\frac{a}{b}$.

5. To divide a line segment into n parts, you can construct a line that makes an acute angle with our original line and use a compass to make n markings on the construction. Then construct a line from the last marking to the endpoint of our original line and construct all the other lines from the each marking to the original line so that all the constructions are parallel.

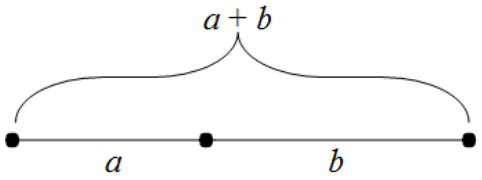


Figure 2: Adding Constructible Lengths

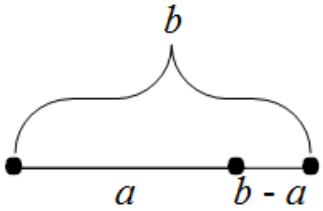


Figure 3: Subtracting Constructible Lengths

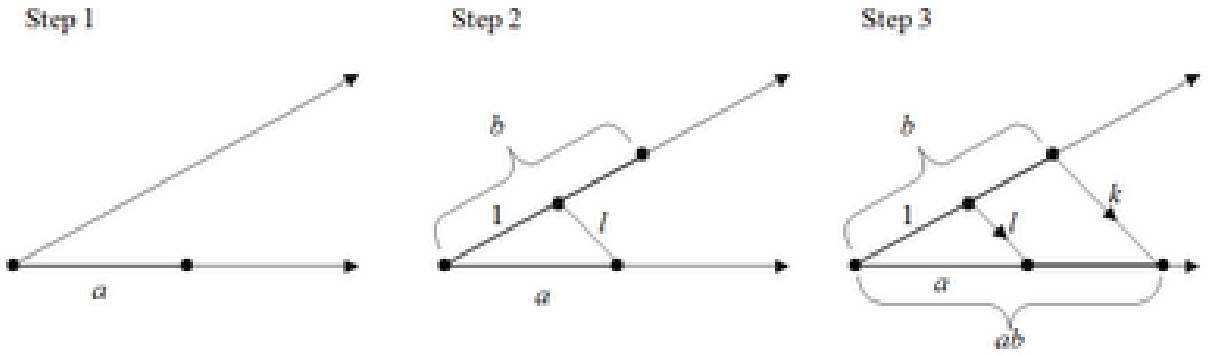


Figure 4: Product of Two Constructible Lengths

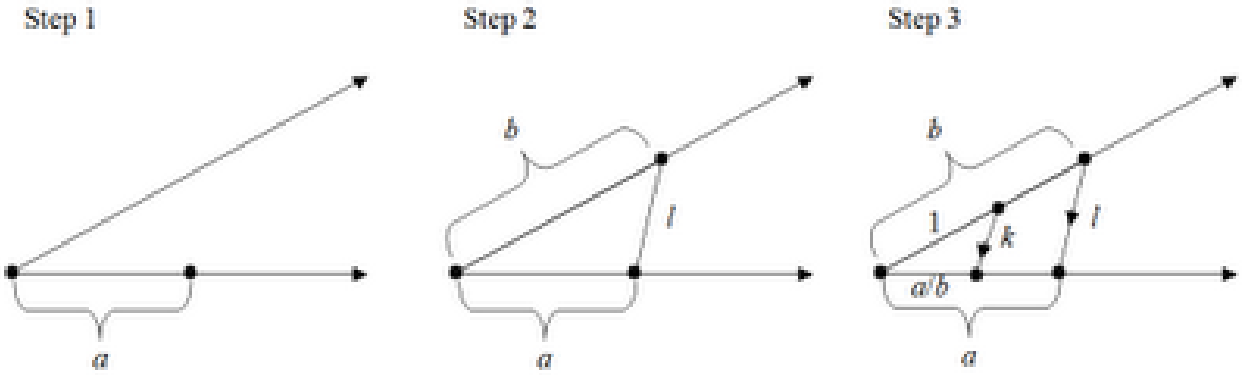


Figure 5: Quotient of Two Constructible Length

3 The Field of Constructible Numbers and Impossible Constructions

Our previous claim that we can perform the basic operations to constructible numbers to form other constructible numbers leads to the question of knowing whether or not a value is constructible. For example, is $\sqrt[3]{2}$ a constructible number? Let's first start with $\sqrt{2}$.

If we construct 2 unit lengths that are perpendicular and share an endpoint, the construction of the diagonal would have a length of $\sqrt{2}$ units because it is the hypotenuse of a right triangle.

In a similar manner, it is possible to construct the square root of any constructible number. In fact, since we are always adding lines and circles when constructing lengths, new constructible numbers are always just solutions to quadratics with coefficients of other constructible numbers. In other words, constructible numbers can always be represented as combinations of the basic 4 operations and square roots applied to rational numbers.

Put another way, the constructible numbers are the smallest extension of Q such that the square root of every constructible number is constructible.. However, you cannot form through the intersections of constructible numbers, the values of cube roots; therefore constructing a line of length $\sqrt[3]{2}$ with only a compass and ruler is impossible.

The impossibility of constructing $\sqrt[3]{2}$ was not proven until the 19th century, along with many other lengths as a constructible number such as π and e. This is simply because no combination of the four basic operations or square roots of any combinations of the four operations is equivalent to a cube root (aside from the cube roots of 0 and 1), and that is how a constructible number is defined. This is also the reason the Ancient Greeks were not able to construct the length to double the volume of a cube using a ruler or compass, one of their major classical problems. (We now know it's impossible, as well as the other problems: trisecting an angle and squaring a circle).

The problem of doubling the cube is the following: given an isometric drawing of a cube, construct an isometric drawing of a cube with double the volume. If s is the side length of the original cube, then clearly, the new cube should have a side length of $s\sqrt[3]{2}$. Therefore, doubling the cube is equivalent to constructing the length $\sqrt[3]{2}$, which is impossible as we just discussed.

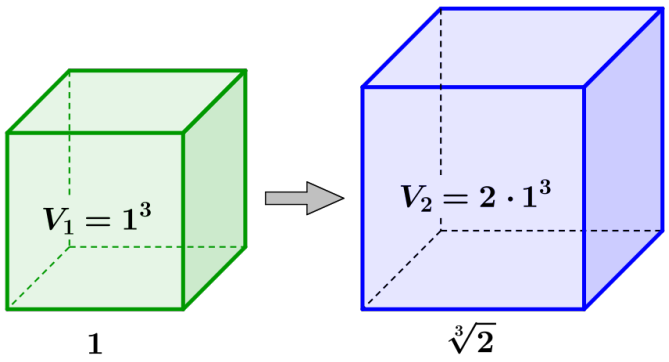


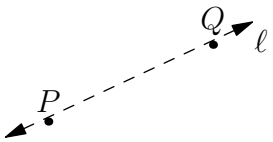
Figure 6: Doubling the Volume of a Cube

4 Huzita-Justin Axioms

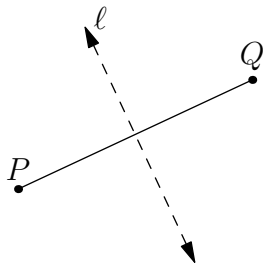
As we shift to constructions by Origami, it is important to note that all ruler and compass constructions can be performed with origami. Origami constructions don't have a strictly defined field like the constructible numbers, but it can be thought of as an extension to the field of constructible numbers that includes values of cube roots, which we will be examined in the following sections. But before that, we introduce the Huzita Justin Axioms.

The Huzita-Justin axioms are a set of origami operations that provide the fundamentals for folds and creases. There are seven axioms in total, where two are special cases, and most of them can be understood intuitively. Note that the construction of lines in these axioms represent the physical folding of paper

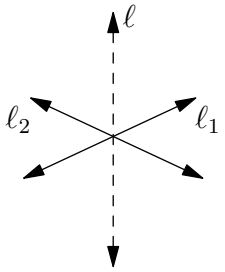
1. Given two points, P and Q, we can construct a line that contains both points.



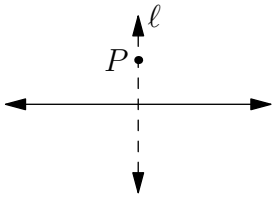
2. Given two points, P and Q, we can construct a line such that P and Q are reflections over that line.



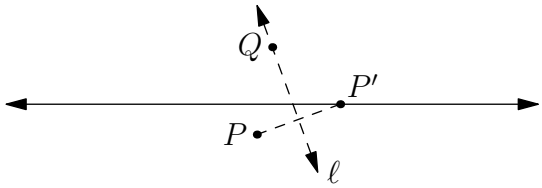
3. Given two lines, ℓ_1 and ℓ_2, we can construct a line such that ℓ_1 and ℓ_2 are reflections over this line.



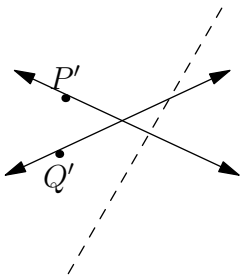
4. Given a point P and a line ℓ, we can fold a line that contains P and is perpendicular to ℓ.



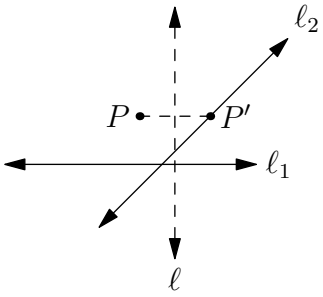
5. Given a line ℓ, and two points, P and Q, not on line ℓ, we can fold a line that places P onto ℓ such that the crease passes through Q.



6. Given two points, P and Q, and two lines, ℓ_1 and ℓ_2, we can fold a line that places P onto ℓ_1 and Q onto ℓ_2



7. Given a point P and two lines, ℓ_1 and ℓ_2, we can fold a line perpendicular to ℓ_1 that places P onto ℓ_2.



The origami axioms are very similar to constructions in geometry, where a crease is equivalent to a line.

5 Constructing a Parabola

One way to define a parabola is with a focus F and a directrix d, which is a line. A parabola is the set of all points that are equidistant from the focus and directrix. A parabola can be constructed through paper folding with if you draw a line to represent the directrix d and a point, preferably centered and somewhat near the directrix, to represent the focus. It is best to use origami paper or thin, transparent paper.

Given a line ℓ and a point P, and you know the distance between ℓ and P. Folding P onto line ℓ near one end of line will form a crease. If you adjust your pull on the flap of paper to place point P further along line ℓ, until you reach the other edge of the line, you will see the creases form a parabola shape that surround point P. This reveals a parabola, where each of the crease lines are the tangents to the parabola.

More simply, you can draw many distinct points on line ℓ and fold the lines that place point P onto each of the points you drew on line ℓ. If you trace the curve formed from all the creases of the tangents, you will see the formation of a U shaped curve: the parabola. The focus of the parabola is point P and the directrix is line ℓ. For parabolas with end behaviors facing up, the parabola is above the directrix, and for parabolas with end behaviors facing down, the parabola is below the directrix.

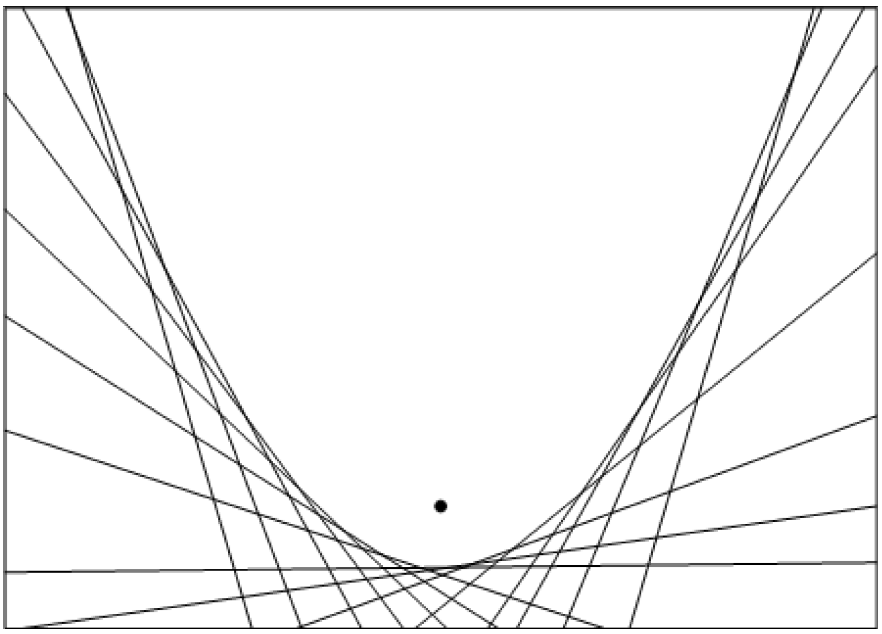


Figure 7: Parabola Formed from Paper Folding

6 Beloch's Big Idea

Margherita Piazzola Beloch is credited for discovering the method to solve any cubic equation using origami. The 6th origami axiom is also known as Beloch's fold, which is key to solving the cubic.

First let's understand how origami can help us solve a quadratic. When we fold a point P to a line m, it will form a tangent to the parabola formed with focus P and directrix m. The crease formed is the perpendicular bisector of the segment connecting P to m, and it is equivalent to finding the intersection of that tangent to the parabola. It can basically be thought of as solving a quadratic.

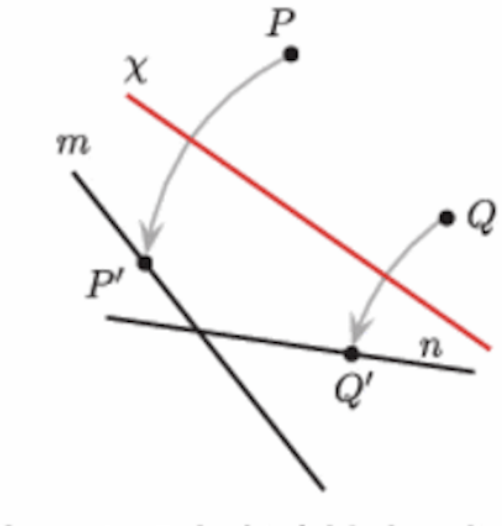


Figure 8: Beloch's Fold

Beloch's fold is important because it forms the crease that is tangent to more than one parabola. Let's say we have one parabola with focus P, directrix m, and a second with focus Q, directrix n. We can use Beloch's fold to fold the line that places P onto m and Q onto n, and the crease made from folding a focus to the directrix of a parabola represents a tangent line to the parabola. The other two tangents aren't as clear to obtain but can be found by folding P to m, and adjusting the fold to be tangent to the second parabola, and by folding Q to n, and adjusting the fold to be tangent to the first parabola. Technically, the ability to find all three tangents of the two parabolas is equivalent to solving a cubic equation.

Extending this concept to solve cubic equations is a little more complex. In order to do so, we need to find the tangent lines of two parabolas. Mathematically, two parabolas can have up to three common tangents.

Suppose we have the parabolas $y = x^2$ and $x = y^2$, and we would like to find the equations of the tangents between them. To find a line $y = mx + b$ to be tangent to the first parabola, we must set them equal to each other and the discriminant should be equal to zero. Therefore, we have

$$\begin{aligned} mx + b &= x^2 \\ x^2 - mx - b &= 0 \\ \Delta = m^2 + 4b &= 0, \end{aligned}$$

so $b = \frac{-m^2}{4}$. We can do the same for the second parabola:

$$\begin{aligned}(mx+b)^2 &= x\\ m^2x^2+2mbx+b^2 &= x\\ m^2x^2+(2mb-1)x+b^2 &= 0\\ \Delta = (2mb-1)^2-4m^2b^2 &= 0\\ 4m^2b^2-4mb+1-4m^2b^2 &= 0\\ -4mb+1 &= 0.\end{aligned}$$

Therefore, we have $b = \frac{1}{4m}$. Equating the two expressions we derived for b, we see that

$$\frac{1}{4m} = \frac{-m^2}{4} \implies m^3+1 = 0.$$

Thus, finding tangent lines to two parabolas reduces to solving a cubic (in the slope of the tangent line). One can change the two parabolas to change the resulting cubic, although we won't show this in full generality here.

Beloch's big idea was able to relate finding the tangents of two parabolas to solving a cubic equation. The solutions of the cubic gives the slopes of the tangent lines, and if we plug them into either of the general tangent equations you can find the y-intercept, and then substituting both values back into $y = mx+b$ form can give you the equation to the tangent line. For our cubic $m^3+1 = 0$, there is only one real solution, $m = -1$, and if we substitute -1 into $b = \frac{1}{4m}$, we get $b = \frac{-1}{4}$. Therefore, $y = x^2$ and $x = y^2$ have only one real tangent line $y = -x - \frac{1}{4}$ (and two non real ones).



Figure 9: The Three Tangent Lines of Two Parabolas

7 Constructing a Cubic Length

Here is a more concrete example of constructing the solution to cubic, known as the Beloch Square.

Given two points, A and B, and two lines, r and s, we can construct a square $WXYZ$ where X lies on line r, and Y lies on line s, and sides WX, ZY, or their extensions, pass through A and B.

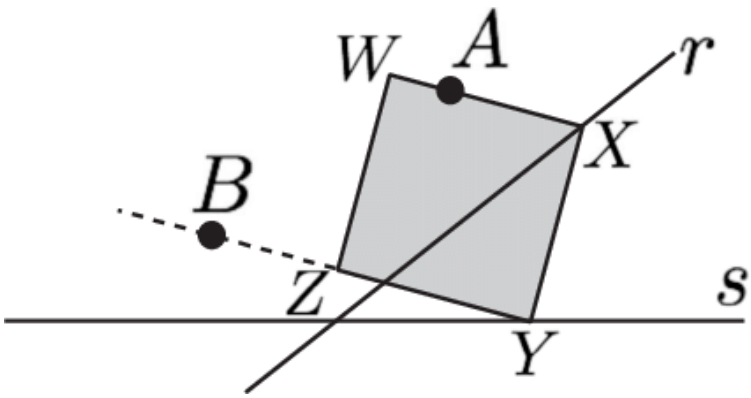


Figure 10: The Beloch Square

To construct this square, we can first construct a line r' parallel to r and twice the distance from point A than r, and do the same thing to construct a line s', using point B and line s. We can then use the Beloch fold to place A onto r' and B onto s', producing points A' and B'. The crease formed from the fold forms a perpendicular bisector to lines AA' and BB', and the intersection form points X and Y. To form the rest of the square, we just have to copy the length XY onto the lines AA' and BB'.

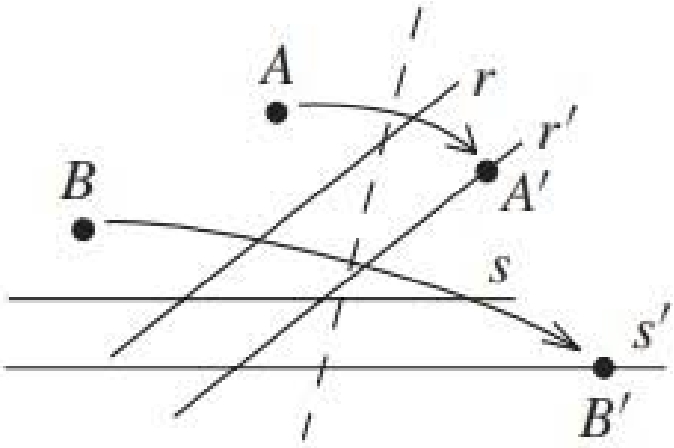


Figure 11: Constructing the Beloch Square

Beloch demonstrated her discovery by using origami to form the length for $\sqrt[3]{2}$. If we place Beloch's square on a coordinate plane, where r is the y-axis, s is the x-axis, O is the origin, A is the point $(-1, 0)$, B is the point $(0, -2)$.

1. $\frac{OX}{OA} = \frac{OY}{OX} = \frac{OB}{OY}$ because the points form similar right triangles.
2. We know $OA = 1$ and $OB = 2$.
3. This leads to the equation $|OX|^3 = 2$.
4. Therefore $OX = 2^{1/3}$ and $X = (0, \sqrt[3]{2})$.

This discovery also found the solution to the classic Greek problem of doubling the volume of cube that required to find $\sqrt[3]{2}$

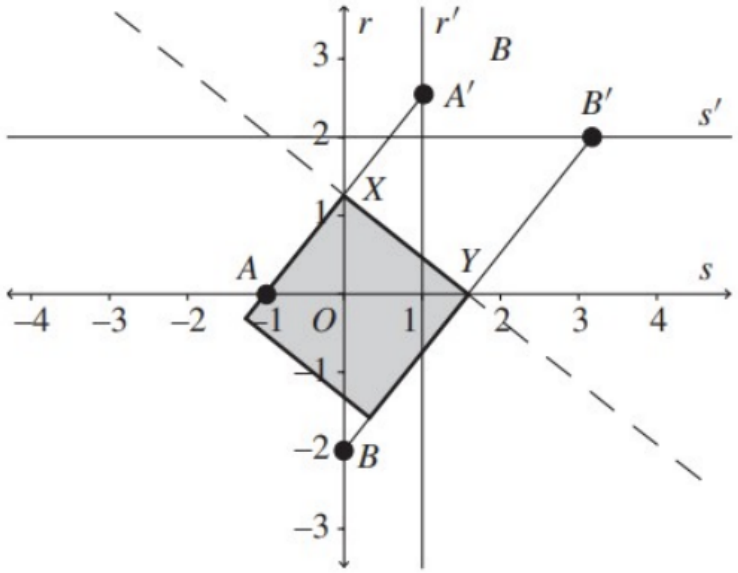


Figure 12: The Beloch Square on a Coordinate Plane

References

1. http://sigmaa.maa.org/mcst/documents/ORIGAMI.pdf
2. https://people.math.ethz.ch/~acannas/Student_Papers/BSc_Theses/2024_bsc_kunz_mathematics_of_origami.pdf
3. https://langorigami.com/article/huzita-justin-axioms/
4. https://www.britannica.com/art/origami
5. http://origametry.net/papers/amer.math.monthly.118.04.307-hull.pdf
6. https://faculty.niu.edu/math_richard/pdfs/ch8.pdf
7. https://en.wikipedia.org/wiki/Constructible_number
8. https://www.math.toronto.edu/mathnet/questionCorner/impossconstruct.html
9. https://quibb.blogspot.com/2013/04/constructible-numbers-and-figures.html
10. https://upload.wikimedia.org/wikipedia/commons/thumb/f/fc/01-W%C3%BCrfelverdoppelung-4.svg/960px-01-W%C3%BCrfelverdoppelung-4.svg.png
11. https://www.cutoutfoldup.com/507-fold-a-parabola.php
12. https://www.researchgate.net/figure/The-Beloch-square_fig2_266329130
13. https://digitalrepository.salemstate.edu/handle/20.500.13013/2977g
14. https://www.themathdoctors.org/tangents-without-calculus/